

Project Documentation — HackMatrix 2026

Team Name

Team *TwinSparks*

Team Leader Name + Contact

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Team Members

- Lakshmi Srujana — Development & Integration Lead; Version Control
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Your Problem Statement

Closing the Pledge-to-Delivery Gap in Community Relief

Domain: Disaster Management / GovTech / Blockchain

Event Name

HackMatrix 2026 – Round 2

Links

- **GitHub Repository:** <https://github.com/lakshmi-srujana/groundwork.git>
- **Demo Video Link:** <https://youtu.be/x9AJqq4IV78>

Project Name / Title

Groundwork — *Because Showing Up Should Count.*

Platform Preview

Groundwork's interface includes:

- **Coordinator Command Matrix** — a dashboard showing active volunteers, verification queue, live relief-tasks, live proof-telemetry feed, and anomaly/defense alerts.
- **Volunteer View** — a personal task list showing pledged deliveries (item, quantity, ward, location) with status tags (Verified / Pending / On Ledger).
- **Pledge a ReliefTask** form — volunteers log item, quantity, ward/location, and optional description before capturing live proof.
- **Live ProofCapture Screen** — camera-only photo capture tied to a specific pledge, submitted for AI verification.

Summary

Groundwork is a platform that lets disaster relief volunteers prove their contributions actually happened. It combines live, camera-only photo capture with GPS metadata, AI-based verification against the pledged task, and an immutable on-chain record — turning informal WhatsApp-style pledges into verifiable, portable proof-of-impact.

Problem Being Solved

In India's disaster relief-ecosystem, over 1,00,000 trained Aapda Mitra volunteers operate across 350 disaster-prone districts, but:

- **No system exists to confirm a pledge was actually fulfilled** — delivery is fundamentally unverifiable today.
- **Photo "proof" is just a gallery upload** with no location or time attached, so fabricated or reused evidence is indistinguishable from real evidence.
- **Even genuine deliveries leave no permanent, checkable record** — reliable volunteers get no credit, and coordinators waste time re-vetting people from scratch during every single crisis.

The current method — a pledge made in a WhatsApp group — creates friction, delay, and risk, because there's no way to confirm anything, and no record survives the crisis to build reusable trust.

USP (Unique Selling Point)

Groundwork replaces "we'll take your word for it" with evidence that would require faking three independent signals at once:

Existing Approach	Groundwork
Proof= WhatsApp photo, unverifiable	Live camera-only capture, GPS-attached at time-of-proof
Verification = none, or manual and slow	AI-verified in real time, with human review only for uncertain cases
Record = a chat message that disappears	Immutable, tamper-evident record on Polygon
Volunteer reputation = doesn't exist	Portable trust profile that follows the volunteer across every crisis
Trust = "we'll take your word for it"	Trust = evidence three separate signals would all have to be faked at once

Key Features

1. **Camera-Only Live Capture** — No gallery uploads, no reused photos. GPS is attached the moment the photo is taken, so evidence can't be faked from a desk.
2. **AI Verification with a Human Backstop** — Gemini Vision checks the photo against the actual pledge. Confident matches auto-verify; uncertain cases are routed to a human coordinator instead of being silently approved or rejected.
3. **Tamper-Evident Proof-of-Fulfillment** — Verified deliveries get an immutable record written to Polygon, linked to evidence stored on IPFS, so no one can quietly edit "verified" to "rejected" after the fact.
4. **Portable Trust Profile** — A volunteer's verified history follows them across every crisis, eliminating repeated re-vetting.
5. **Coordinator Command Matrix** — A live dashboard for NGOs/district coordinators to track active volunteers, verification queues, and relief-task status in real time.

How it works (flow)

1. **Volunteer Pledge** — Volunteer logs item, quantity, ward, and location.
2. **Live Camera Capture** — Camera-only photo with EXIF geotag (no gallery uploads).
3. **AI Verification Oracle** — Face detection, geotag-vs-ward check, item/quantity match; flags or verifies.
4. **Proof-of-Fulfillment Minted** — Cryptographic hash stored on-chain, immutable.
5. **Evidence Stored Decentrally** — Photo + metadata pinned to IPFS, linked to the proof hash.
6. **Volunteer Trust Profile Updated** — Portable record updates; org dashboard unlocked.

Tech Stack

- **Frontend:** React, Next.js 14, Tailwind CSS
- **Backend:** FastAPI (Python) — handles verification and blockchain calls
- **Database:** Supabase (PostgreSQL) for volunteers, tasks, submissions; Supabase Realtime for live coordinator updates
- **AI/ML:** Gemini Vision API for photo-to-pledge verification
- **Blockchain:** Polygon + Thirdweb for on-chain verification records; IPFS / Web3.Storage for decentralized evidence storage
- **Deployment:** *[add actual host — e.g. Vercel for frontend, Railway/Render for FastAPI]*

Feasibility

Technical

- Gemini Vision API — already available and integrated for verification
- Polygon + Thirdweb — on-chain writes without custom chain infrastructure
- Supabase — auth, database, and real-time updates in one service

Operational

- Runs in a phone browser — no app install, no extra hardware required
- Deployment: web app, live at pilot scale

Hackathon MVP Pledge → live photo + GPS → AI verification → verified record on Polygon
→ coordinator dashboard

Impact

- **Who benefits?** Aapda Mitra volunteers and the NGOs/district coordinators who deploy them.
- **What improves?** Volunteers get portable proof-of-contribution; coordinators get a verified roster instead of a WhatsApp thread.
- **Why does it matter?** Faster trust means faster coordination — organisations stop re-vetting from zero during every single disaster.

Future Scope

Next MVP

- Real volunteer user testing
- Reliability hardening on the AI verification oracle
- Security review of the smart contract
- UX refinement based on coordinator feedback

Later

- Real-world pilot deployment with an NGO or district disaster-management cell
- Analytics on verification accuracy
- Integrations with existing relief-coordination tools

Scale

- Expansion to more volunteers and more disaster types beyond initial pilot districts
- NDMA (National Disaster Management Authority) integration
- A national volunteer trust score



